IT Italiano · GB **English**




# 🎮 ichosynth — Usage Manual

### How to play the sampler-sequencer you *draw*

You draw music on a 16×16 grid with **4 knobs** and **3 buttons**. No computer, no menus to memorize: turn, press, listen.

Level Beginner · Build 4 encoders + 3 buttons · firmware TŒRN · SP_ · See also Build Manual

> 🎧 **You don't need a computer to play**: your **ichosynth** generates everything on its own. Plug in your **headphones**, power it via USB, and off you go.

> 🔌 **Connections — getting sound out (and in)**: the instrument has two audio jacks.
>
> - **Headphones** — the audio board's **3.5 mm stereo** jack: this is the output (TŒRN pans the voices in stereo). You can also feed an amplifier or mixer from it with a 3.5 mm cable.

- **Line In** — 1× **6.35 mm (1/4") MONO** jack. Feed in an instrument, another synth, a field recorder or a mixer send to **record/sample it** (see ch. 16).

ℹ️ **This is the real TŒRN.** ichosynth runs the full **TŒRN firmware** (by SP_/soundpauli, toern.live) on a Teensy 4.1, built with cheap hand-soldered parts: **4 KY-040 encoders**, **3 tact switches** and an **SSD1306 OLED**. Where the original TŒRN told you the state with the glowing colour of its encoder rings, our build shows it as plain text on the **OLED**.

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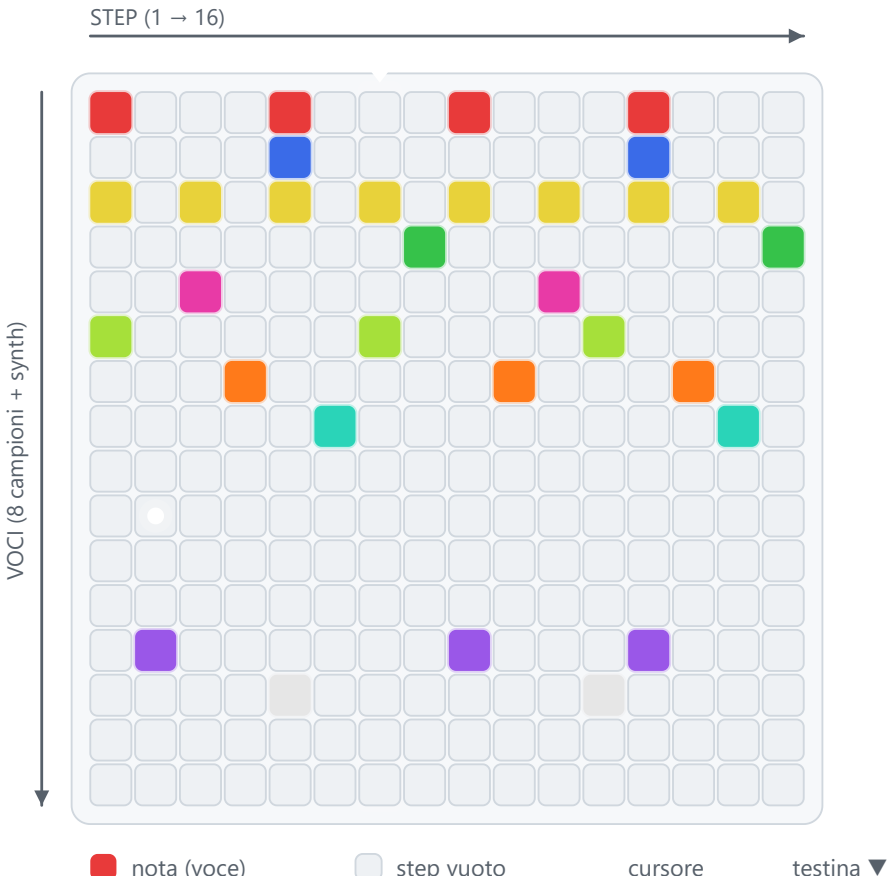

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## 1 · The concept in 30 seconds

The **16×16 grid** is your music sheet. A playback "playhead" runs from left to right: every column it touches plays the notes you've put there.

## La griglia 16×16 = il tuo foglio musicale

Righe = voci (colori) · Colonne = step · la testina suona colonna per colonna



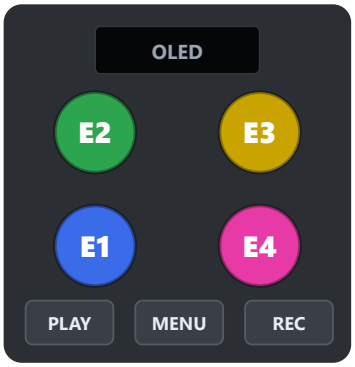

- The **columns** (left→right) are the **16 steps** of one bar (a page can be chained to **32×16**).
- Each **row** is a **voice** (a sample or a synth), identified by a **color**.
- Several pages in a row make up a **pattern**; several patterns make up a **song**.

> 💡 Basic flow: **draw notes** → **press PLAY (B1)** → **loop**. You change samples, BPM and volume on the fly, without stopping.

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## 2 · The hardware: 4 knobs + 3 buttons

Next to the grid there are **4 encoders** — **E1 E2 E3 E4** — and below them **3 buttons**: **B1 PLAY**, **B2 MENU**, **B3 REC**. Every encoder both **turns** and **pushes** (click). The **OLED** sits on top.



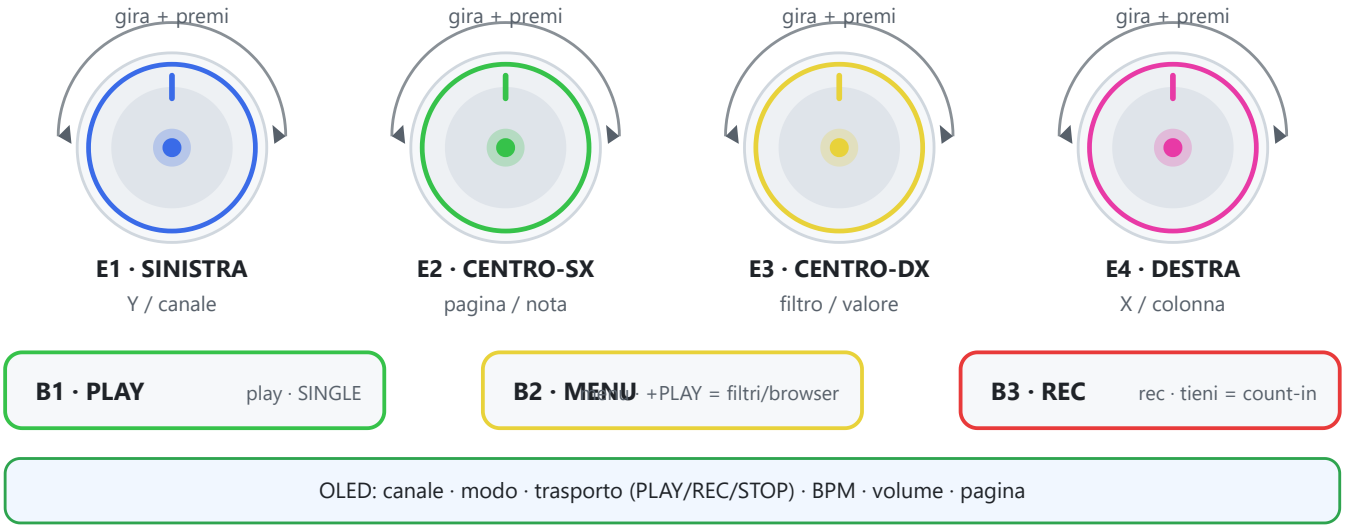

With the 3D-printed case the controls sit like this:

- **E2 E3** on top — page and quick filter;
- **E1 E4** at the bottom, nearest the hand — the **cursor** (E1 = up/down, E4 = left/right);
- **PLAY · MENU · REC** in a row: PLAY and MENU side by side for the two-finger combo.

Without the case lay them out as you like: what counts is the encoder's **number**, not its position.

## I comandi: 4 encoder + 3 pulsanti

4 manopole (gira e premi) + 3 tact switch. Lo stato si legge sull'OLED.




The gestures the firmware understands:

| Gesture | What it means |
|---|---|
| **Turn** (E1–E4) | change the value / move the cursor in the current context |
| **Short click** (push an encoder) | confirm / act (the meaning depends on the mode) |
| **Long press** (hold an encoder) | a second action (e.g. E1 hold = mute/subpattern) |
| **Tap a button** (B1/B2/B3) | the button's main function |
| **Hold a button** | the button's held function (e.g. B3 held = record) |
| **Combo B1+B2** | a shortcut — **the order you press them matters** (see ch. 11 & 12) |

> 💡 There is **no double-click**: the KY-040 encoders don't need one. Every action is a turn, a click, a hold, a button tap or a two-button combo.

What each encoder does changes with the mode. Here is the overview (details in the relevant chapters):

| Mode | E1 turn | E2 turn | E3 turn | E4 turn |
|---|---|---|---|---|
| **DRAW** | Y / note (row) | page | quick channel filter | X / column |
| **SINGLE** | channel | note (pitch) | — | X / column |
| **FILTER** | slider 1 | slider 2 | slider 3 | slider 4 |
| **MENU** | — | value | value | navigate menu pages |
| **VELOCITY** | velocity | probability | channel volume | condition / timing |
| **SONG** | — | pattern | — | position (1–64) |

The most useful encoder **clicks** (in DRAW/SINGLE):

- **E3 click** = **Play / Pause**.
- **E1 long-press** = enter mute / subpattern (release to restore).
- **E1 click @ row 16** (in SINGLE) = NOTE SHIFT.
- **E4 click** = confirm / enter a sub-menu.
- **E1 click** (in MENU) = back / exit.

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## 3 · Reading the grid and the OLED

- **Rows** = the voices (8 sample voices + 3 synth voices), each with its own **color** (see ch. 13).
- **Columns** = the 16 steps of the current page.
- **Top row (status)**: page indicators and status lights (copy active, etc.). During Play the page indicators turn **green**.
- **Play playhead**: the highlighted column that advances as you play (the ▼ in the image above).

> 📟 The original TŒRN showed the live state through the **colour of the encoder rings**. Our build has no RGB rings, so the **OLED** shows the same information in plain text: **current channel, mode, transport (PLAY / REC / STOP), BPM, volume and page**. Glance at it any time to know exactly where you are.

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## 4 · The 3 buttons (B1 B2 B3)

The three tact switches are the heart of the transport and navigation:

| Button | Main function | Also does… |
|---|---|---|
| **B1 · PLAY** | start playback; toggle **SINGLE**; exit other modes back to **DRAW** | **held at power-on** = clear the RAM (fresh start) |
| **B2 · MENU** | enter / exit the **Menu** and the sub-modes | exits a sub-mode back to DRAW/SINGLE |
| **B3 · REC** | **record** (tap or hold) | **PAUSE** during playback · **tap-tempo BPM** · in REC mode **ON1**, hold **>300 ms** = **count-in** |

> 💡 **PLAY is B1** (a single tap), not an encoder gesture. **PAUSE is B3** while a song is running.

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## 5 · DRAW mode (drawing)

This is the main screen, the default one, where you create patterns.

| Action | Gesture |
|---|---|
| 🧭 **Move the cursor** | turn **E1** (up/down, row/note) and **E4** (left/right, column) |
| 📄 **Change page** | turn **E2** |
| 🖍️ **Add / change a note** | push on the cursor's spot (you hear it right away); pressing again on a note cycles the voice |
| ▶️ **Play / Pause** | **E3 click** (Pause during play is also **B3**) |
| 🎚️ **Quick filter the channel** | turn **E3** to soften/brighten the voice under the cursor on the fly |
| 🔇 **Mute / subpattern** | **hold E1** (release to restore) |

> 💡 E3 is the "transport" knob in DRAW: **click** it for Play/Pause, **turn** it to nudge the current channel's filter without leaving the grid.

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## 6 · Pages, patterns and subpatterns

- The grid shows **one page** (16 steps) at a time; turn **E2** to move between pages.
- Pages with notes are played in sequence on a loop to form a **pattern**. Patterns are chained in SONG mode.
- **Subpatterns**: **hold E1** to drop into a momentary variation of the current pattern, then release to snap back — great for live fills and breaks.
- **Note copy / paste and note-shift** let you move and duplicate steps; the active copy/note-shift state shows on the grid's top row (and on the OLED).

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## 7 · Mute (silencing voices)

- Put the cursor on a voice and **hold E1** to mute it / drop into its subpattern; release to restore.
- Muted channels are shown **on the grid itself**, so you can always see what's silent at a glance.

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## 8 · Volume and BPM

- **Volume** and **BPM** are always readable on the **OLED**.
- **Tap-tempo**: tap **B3 (REC)** in time to set the **BPM** by ear.
- The per-channel **volume** is adjusted in VELOCITY mode with **E3**.

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## 9 · Velocity, probability and conditions

TŒRN gives every step more than just on/off. In **VELOCITY** mode the four knobs become per-step controls:

| Knob | Controls |
|---|---|
| **E1** | **velocity** (how loud the step is) |
| **E2** | **probability** (chance the step fires) |
| **E3** | **channel volume** |
| **E4** | **condition / timing** (e.g. play every Nth pass, micro-timing) |

Use **B2 (MENU)** to step through the sub-modes; **B1 (PLAY)** returns you to DRAW.

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## 10 · SINGLE mode (one voice only)

Useful for working in detail on one sample (e.g. a melody across several pitches).

- **Enter / exit SINGLE**: tap **B1 (PLAY)** to toggle SINGLE for the focused voice.
- In SINGLE: **E1 turn** = pick the **channel**, **E2 turn** = the **note (pitch)**, **E4 turn** = the **column (X)**.
- The same drawing/deleting gestures from DRAW apply, but only to the selected voice.

### Note Shift (in SINGLE)

- Put the cursor on row 16 and **click E1** to enter **NOTE SHIFT**, then move the notes with the knobs.

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## 11 · FILTER mode and per-voice DSP

ichosynth has the full TŒRN per-voice DSP: a **lowpass filter**, **reverb**, **bitcrusher**, **detune**, **octave**, and a **Moog ladder** on the synth voices.

- **Enter FILTER mode**: press **B2 + B1 together, with B2 first** (B2-first = FILTER MODE). This works on the channels that have filters (1–8, 11, 13–14).
- In FILTER mode the four knobs become **four sliders** (**E1 E2 E3 E4**) for the DSP of the selected voice. The OLED shows the selected filter and its value.
- Exit with **B2 (MENU)** or **B1 (PLAY)** back to DRAW.

> 💡 For a fast, no-mode tweak you can also just **turn E3 in DRAW** to filter the channel under the cursor. FILTER mode is for dialling in the full set of effects per voice.

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## 12 · Changing the sample (Sample Browser)

To assign a different WAV to one of the 8 sample voices:

1. **Open the Sample Browser**: press **B1 + B2 together, with B1 first** (or both at once). This enters SET_WAV for channels 1–8.
2. **Navigate** with the knobs: change **folder** and **sample** (browse the SD); you can preview the **length** and **waveform**, and set **seek / length / reverse**.
3. **Confirm** with **E4 click** to load the selected sample onto the voice.
4. **Exit** with **B2 (MENU)**.

> 📁 Browser samples live on the microSD as `samples/<category>/<name>.wav` (up to 256 per folder; see the build manual). WAVs must be mono, 16-bit, 44.1 kHz — `wavmaker` prepares them for you.

> ⏱️ **Order matters in the combo**: **B2 first** → FILTER MODE; **B1 first (or simultaneous)** → SAMPLE BROWSER. There is a short 350 ms cooldown between combos.

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## 13 · Voice colors and the synth voices

Each voice has a fixed color (defined in `colors.h`):

### Legenda colori delle voci

Ogni riga della griglia è una voce, identificata da un colore (da colors.h).

| | | |
|---|---|---|
| **Voce 1** | rosso | voce campione |
| **Voce 2** | blu | voce campione |
| **Voce 3** | giallo | voce campione |
| **Voce 4** | verde | voce campione |
| **Voce 5** | magenta | voce campione |
| **Voce 6** | verde-lime | voce campione |
| **Voce 7** | arancione | voce campione |
| **Voce 8** | turchese | voce campione |
| **Voce 13** | viola | synth (sawtooth) |
| **Voce 14** | bianco | synth (square) |

There are **8 sample voices** plus **3 synth voices**. The synth voices play internally generated waves and follow the pitches of a scale; they have their own **Moog ladder** filter (set in FILTER mode). TŒRN is **polyphonic**, so several voices sound together.

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## 14 · Sample packs

A "sample pack" is a complete set of voices saved on the SD: recall a whole kit on the fly.

- Open the Sample Browser (ch. 12) and use the pack controls to **load** a numbered pack onto the 8 sample voices.
- Packs let you swap an entire kit between songs without rebuilding it voice by voice.

> 📁 On the SD a pack is a numbered folder `0` … `99` holding 8 voices `1.wav` … `8.wav` (pack 0 is the factory kit). The ready-made SD in `_SDCARD/` has 6; use `wavmaker` to make more.

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## 15 · Saving and loading (Menu)

- **Enter the Menu**: tap **B2 (MENU)**.
- Navigate the menu pages with **E4** (turn), adjust values with **E2 / E3**, and **confirm / enter a sub-menu with E4 click**; **E1 click = back**.
- From the Menu you **save** and **load** songs to/from the microSD.
- **Exit** the Menu with **B2 (MENU)** again, or **B1 (PLAY)** to jump straight back to DRAW.

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## 16 · Live recording (REC)

ichosynth records audio straight from its input into the current channel.

> 🎙️ **The Line In jack is the default record source.** Connect an instrument, another synth, a field recorder or a mixer send to the **6.35 mm (1/4") MONO Line In** jack and you can sample it straight in — no menu setup needed (the take is a mono sample). See the **Connections** note at the top of the manual for the jacks.

1. Choose the **channel** to record into.
2. **Hold B3 (REC)** to record from the input: in this build it is **LINE IN** (you can switch to MIC from the menu; the choice is remembered).
3. If the recording mode is **ON1**, hold **B3 > 300 ms**: first a 4-beat **count-in** at the current BPM, then recording starts and stops on the beat.
4. Release **B3** to stop; the take is saved to the SD and loaded onto the channel, so it survives a reboot.

> 💡 **B3 is multi-purpose**: a quick tap is **tap-tempo** / **PAUSE during play**; a hold is **record**. The OLED's transport readout shows **REC** while you're recording.

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## 17 · SONG mode

SONG mode chains your patterns into a full arrangement.

- In SONG mode: **E2 turn** = choose the **pattern**, **E4 turn** = the **position** in the song (1–64).
- Build the sequence of patterns, then **PLAY (B1)** runs the whole song.
- Mute states and the song position are shown **on the grid**, so you can follow the arrangement live.

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## 18 · MIDI and tap-tempo

- **MIDI**: TŒRN speaks **DIN MIDI** (clock and notes) on a Teensy serial port (**pin 34 = IN, 35 = OUT**). It needs a small MIDI circuit (optocoupler + DIN sockets), **not part of the base build**; the firmware does not handle MIDI over USB.
- **Tap-tempo**: tap **B3 (REC)** in time to set the BPM by feel — no menu needed.

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## 19 · Mode & command map

From DRAW (the main screen) you reach everything with knobs, buttons and the two combos:

## Mappa delle modalità — cosa fa ogni encoder

I 4 encoder cambiano funzione col modo. I 3 pulsanti restano costanti.

| | E1 ruota | E2 ruota | E3 ruota | E4 ruota |
|---|---|---|---|---|
| **DRAW** | Y / nota | pagina | filtro rapido | X / colonna |
| **SINGLE** | canale | nota (pitch) | — | X / colonna |
| **FILTER** | slider 1 | slider 2 | slider 3 | slider 4 |
| **MENU** | — | valore | valore | naviga pagine |
| **VELOCITY** | velocity | probabilità | volume canale | condizione |
| **SONG** | — | pattern | — | posizione |

I 3 pulsanti (sempre attivi):

**B1 PLAY**
play/pausa · SINGLE · esci

**B2 MENU**
menu · +PLAY = FILTER / BROWSER

**B3 REC**
rec · tap-tempo · tieni = count-in

Legend: **E1–E4** = the four knobs (turn or click) · **B1/B2/B3** = the three buttons · "click" = short press · "hold" = long press · "combo" = two buttons together (order matters).

| You want to… | Gesture |
|---|---|
| Move cursor up/down (row/note) | turn **E1** |
| Move cursor left/right (column) | turn **E4** |
| Change page | turn **E2** |
| Add / change a note | push on the cursor's spot |
| **Play / Pause** | **E3 click** (Pause also **B3**) |
| Quick-filter the current channel | turn **E3** (in DRAW) |
| Mute / subpattern (current voice) | **hold E1** |
| **PLAY** (transport) | **B1** |
| **MENU** (save/load, settings) | **B2** |
| **REC** / pause / tap-tempo | **B3** |
| Clear RAM (fresh start) | **hold B1 at power-on** |
| Enter/exit **SINGLE** | **B1** (toggle) |
| **Note Shift** (in Single) | **E1 click @ row 16** |
| **VELOCITY** (velocity/prob/vol/condition) | **B2** to reach it → **E1/E2/E3/E4** |
| **FILTER mode** (per-voice DSP) | **B2 + B1, B2 first** → sliders on **E1–E4** |
| **Sample Browser** (set WAV) | **B1 + B2, B1 first / together** → confirm **E4 click** |
| **Menu** confirm / enter sub-menu | **E4 click** · back = **E1 click** |
| Recording **count-in** (REC mode ON1) | **hold B3 > 300 ms** |

> ⚠️ **The two combos differ only by order**: **B2 first** → FILTER mode; **B1 first (or both at once)** → Sample Browser. Wait ~350 ms between combos.

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## 20 · Common problems

| Symptom | Fix |
|---|---|
| 🔇 **I don't hear anything** | check the volume/channel volume (OLED), that the voice isn't muted (**hold E1**), and that the sample exists on the SD in the right format |
| ▶️ **It won't start** | Play is **B1** (or **E3 click**); Pause during play is **B3** |
| ↩️ **A knob goes the wrong way** | it's a CLK/DT swap from the wiring stage (see the build manual) |
| 💾 **noSD icon** | the SD goes in the **Teensy's** slot, FAT32 |
| 🚫 **The samples won't play** | wrong SD path/structure, or WAV not mono/16-bit/44.1 kHz → use `wavmaker` |
| 😳 **A combo opened the wrong mode** | mind the order: **B2 first** = FILTER, **B1 first** = Sample Browser; wait ~350 ms and retry |
| 💾 **I want a fresh start** | **hold B1 at power-on** to clear the RAM; save songs to the SD from the Menu (B2) |
| ⏱️ **Sample length** | long samples with continuous looping are supported |

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Have fun! 🎶

*ichosynth runs **TŒRN** by SP_ (soundpauli) · toern.live · on a Teensy 4.1 with 4 KY-040 encoders, 3 tact switches and an SSD1306 OLED.*